

(Task 1 of 4) In this tutorial, we will review previously learned content.

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(Task 2 of 4) What is the result of running this program?

Python | Python [Run 

Lispy

```
i = j * 3    (defvar i (* j 3))
j = 2       (defvar j 2)
print(i)
print(j)    i
           j
```

error

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You predicted the output correctly 🎉🎉🎉

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The first definition tries to bind `i` to the value of `j * 3`. To evaluate `j * 3`, we need the value of `j`. But `j` is not bound to a value at that moment.

Click [here](#) to run this program in the Stacker.

(Task 3 of 4) What is the result of running this program?

Python | 

Python [Run 

Scala 3

```
a = 3
def foo(b):
    def bar():
        c = 6
        return a + b + c
    return bar()
print(foo(4) + 2)

val a = 3
def foo(b : Int) =
    def bar =
        val c = 6
        a + b + c
    bar
println(foo(4) + 2)
```

error

5

The answer is 15.

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Textual explanation

You might think `a + b + c` can't refer to `a` and `b`. However, `a + b + c` appears in the body of `bar`, which is a sub-block of the body of `foo`, which is a sub-block of the top-level block. So, it can refer to the `a` defined in the top-level block and the `b` defined in `foo`. In SMoL, variable references follow the hierarchical structure of blocks.

<https://n-jnvgavl4buxc3xao7d4lq2zxreghpugshyqyu5y-0lu-script.googleusercontent.com/userCodeAppPanel>

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Click [here](#) to run this program in the Stacker.

What is the result of running this program?

Python | 

Python [Run 

JavaScript

```
aa = 3
def abc(bb):
    def h():
        cc = 2
        return aa * bb * cc
    return h()
print(abc(4) * 1)
```

```
let aa = 3;
function abc(bb) {
    function h() {
        let cc = 2;
        return aa * bb * cc;
    }
    return h();
}
console.log(abc(4) * 1);
```

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8

You predicted the output correctly 🎉🎉🎉

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(Task 4 of 4) What is the result of running this program?

Python | 

Python [Run 

JavaScript

```
def k(a):
    b = 1
    return a + b
print(k(3) + b)

function k(a) {
    let b = 1;
    return a + b;
}
console.log(k(3) + b);
```

error

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You predicted the output correctly 🎉🎉🎉

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`k(3) + b` is evaluated in the top-level block, where `b` is not defined. So, this program errors.

Click [here](#) to run this program in the Stacker.

You have finished this tutorial 🎉🎉🎉

Please print the finished tutorial to a PDF file so you can review the content in the future. **Your instructor (if any) might require you to submit the PDF.**

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